



Modelling Financial Markets

Chapter 7: Value-at-Risk and Expected Shortfall



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIS: Bank for International Settlements
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CF: Cornish–Fisher (expansion)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ EVT: Extreme Value Theory
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FX: Foreign eXchange
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GEV: Generalised Extreme Value (distribution)
- ▣ GPD: Generalised Pareto Distribution
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ QML: Quasi-Maximum Likelihood
- ▣ SAV: Symmetric Absolute Value (CAViaR specification)
- ▣ SE: Standard Error
- ▣ US: United States

Acronyms used in this material (2/2)

- **XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)**

Outline

Motivation

Value-at-Risk and Expected Shortfall

Coherent Risk Measures

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The Horizon: Square-Root-of-Time

Portfolio VaR and Risk Contributions

Extreme Value Theory

Liquidity and Regulation

AI for Scientific Discovery

Summary

References

How Much Can We Lose? RiskMetrics and VaR



J.P. Morgan & Co. building, 23 Wall Street, New York

Photo: Beyond My Ken (2012); CC BY-SA 4.0; Wikimedia

[Commons](#)

- Tomorrow's portfolio value is random; management and regulators need **one number**: a loss exceeded only rarely, to set limits and **capital**
- J.P. Morgan launched **RiskMetrics** in October 1994: a free methodology and data set for measuring market risk [J.P. Morgan/Reuters \(1996\)](#)
 - ▶ Volatilities and correlations for many markets, updated daily
- Its Technical Document (4th ed., 1996) became the reference for VaR; the same year the Basel Committee let banks use internal VaR models for capital [Basel Committee \(1996\)](#)
- Textbooks: [Franke, Härdle & Hafner \(2019\)](#), [McNeil, Frey & Embrechts \(2015\)](#), [Danielsson \(2011\)](#)

Data Used in This Chapter

Series	Type	Period	Days
S&P 500	US equity index	2000–2026	6 717
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
EUR/RON	BNR reference rate	2005–2026	5 352
Gold	USD per troy ounce, weekdays	2000–2026	6 956
Portfolio	SPY, TLT, GLD (ETFs) and Bitcoin	2014–2026	3 017

- Daily closing values up to 18 September 2026; daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- Each series on its own calendar; the portfolio uses the days on which all four prices exist
- ETFs: prices adjusted for dividends; SPY tracks the S&P 500, TLT long US Treasury bonds, GLD gold

Losses, Horizons and Tail Probabilities

- Return over one day: $X_{t+1} = r_{t+1}$, the log return in %; **loss** $L_{t+1} = -X_{t+1}$, a **log loss**
 - ▶ Positive L = the position lost money; negative L = a gain
 - ▶ In money, with V_t the value of the position: $\text{loss} = V_t(1 - e^{-L_{t+1}/100}) \approx V_t L_{t+1}/100$ for small L ; with simple returns R , $V_t(-R_{t+1})/100$ is exact
 - ▶ VaR converts exactly through this monotone map; ES must be converted scenario by scenario, then averaged
- **Horizon** h : the number of days over which the loss is measured (1 day, 10 days)
- **Level** α : the **tail probability**, typically $\alpha = 1\%$ or 2.5% : “VaR 1%”, “ES 2.5%”
 - ▶ For continuous losses, α is the probability of exceeding VaR; ES averages the losses in that tail
- **Sign convention in this course**: VaR and ES are **positive** numbers when they are losses

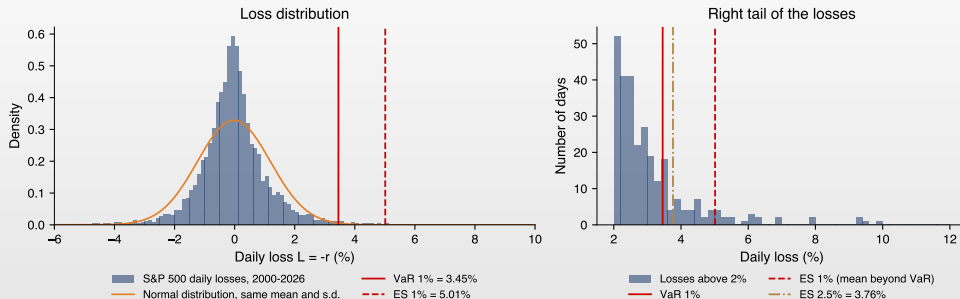
Definition: Value-at-Risk

- **VaR** at level α : the loss that is exceeded with probability α (exactly for continuous L , at most α in general)
 - ▶ $\text{VaR}_\alpha(X) = -\inf\{x \in \mathbb{R} : P(X \leq x) > \alpha\} = -q_\alpha(X)$, minus the α -quantile of the return
 - ▶ Equivalently, for the loss $L = -X$ with a continuous distribution: $\text{VaR}_\alpha = q_{1-\alpha}(L)$
- Reading of VaR 1%: “with probability 1%, tomorrow’s loss will exceed $\text{VaR}_{1\%}$ ”
- What VaR does **not** say: how large the loss is when it exceeds VaR
 - ▶ Two portfolios with the same VaR can have very different tails

Definition: Expected Shortfall

- **Expected Shortfall:** the average of all quantiles in the tail of probability α
 - ▶
$$\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du = \frac{1}{\alpha} \int_0^\alpha \text{VaR}_u(X) du$$
- For a continuous distribution: $\text{ES}_\alpha(X) = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$, the mean loss on the worst α share of days
- Always $\text{ES}_\alpha \geq \text{VaR}_\alpha$: ES looks **into** the tail, VaR only at its entrance
- Other names: Conditional VaR (CVaR), Tail VaR, Average VaR [Rockafellar & Uryasev \(2002\)](#), [Acerbi & Tasche \(2002\)](#)
- Historical estimate: the mean of the observed losses at or above the empirical VaR, a plug-in approximation of the integral (exact when $n\alpha$ is an integer)

VaR and ES on the S&P 500



□ Left: daily losses 2000–2026 and the Normal distribution with the same mean and standard deviation; right: the tail

Reading the Loss Distribution

- ▣ Historical $\text{VaR}_{1\%} = 3.45\%$: 68 of 6 717 days (1.01%) had a larger loss
 - ▶ $\text{ES}_{1\%} = 5.01\%$: on those days the average loss was 5.01%
- ▣ $\text{VaR}_{2.5\%} = 2.51\%$ and $\text{ES}_{2.5\%} = 3.76\%$: ES 2.5% exceeds VaR 1%
- ▣ The Normal distribution fits the centre but is far too thin in the tail (Chapter 1): heavy tails
- ▣ Largest loss: 12.77% on 2020-03-16 (COVID-19 crash), about ten standard deviations

The Normal Distribution: ES in One Line

- $X \sim N(\mu, \sigma^2)$, $z_\alpha = \Phi^{-1}(\alpha) < 0$, with Φ , φ the standard Normal distribution and density functions
 - ▶ $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$; since $\varphi'(u) = -u\varphi(u)$, $\int_{-\infty}^z u\varphi(u) du = -\varphi(z)$
 - ▶ $\text{ES}_\alpha = -\mu - \frac{\sigma}{\alpha} \int_{-\infty}^{z_\alpha} u\varphi(u) du = -\mu + \sigma \frac{\varphi(z_\alpha)}{\alpha}$
- Constants: $-z_{1\%} = 2.326$, $\varphi(z_{1\%})/0.01 = 2.665$; $-z_{2.5\%} = 1.960$, $\varphi(z_{2.5\%})/0.025 = 2.338$
- S&P 500 ($\hat{\mu} = 0.025\%$, $\hat{\sigma} = 1.21\%$): Normal $\text{VaR}_{1\%} = 2.80\%$ vs historical 3.45%
 - ▶ Too low by a factor of 1.24; under the Normal distribution VaR and ES carry no information beyond σ

Worked Example: the Student- t Distribution

- $X = m + s T$ with $T \sim t_\nu$ (location m , scale s , degrees of freedom ν); $t_\alpha = t_\nu^{-1}(\alpha) < 0$, g_ν the density
 - ▶ $\text{VaR}_\alpha = -(m + s t_\alpha)$, $\text{ES}_\alpha = -m + s \frac{g_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$ (for $\nu > 1$)
- Variance $s^2 \nu / (\nu - 2)$ exists only for $\nu > 2$; smaller ν = heavier tails
- Maximum likelihood fit to S&P 500 returns: $\hat{\nu} = 2.67$
 - ▶ $\text{VaR}_{1\%} = 3.46\%$ (historical 3.45%), $\text{ES}_{1\%} = 5.71\%$ (historical 5.01%)
- The t distribution fixes the VaR, but with $\hat{\nu} < 3$ it overstates ES: one parameter must fit both the centre and the tail

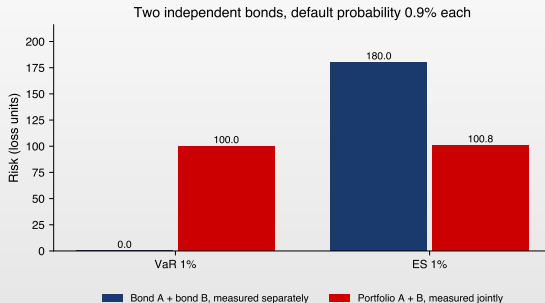
Coherent Risk Measures: Axioms and Why They Matter

- A **risk measure** ρ maps a loss L to capital; it is **coherent** if Artzner, Delbaen, Eber & Heath (1999)
 - ▶ monotone: $L_1 \leq L_2 \Rightarrow \rho(L_1) \leq \rho(L_2)$; translation invariant: $\rho(L + c) = \rho(L) + c$
 - ▶ positively homogeneous: $\rho(\lambda L) = \lambda \rho(L)$, $\lambda \geq 0$; subadditive: $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$
- **Limits**: by subadditivity, bank-wide risk does not exceed the sum of the desk risk limits, $\sum_i \rho_i$; **no regulatory arbitrage** by splitting a firm
- **Optimisation**: homogeneity + subadditivity $\Rightarrow$ convexity in the portfolio weights
 - ▶ Every local minimum is global; uniqueness would need **strict** convexity, which scenario-based ES lacks (it is piecewise linear)
- VaR satisfies the first three axioms, but **not** subadditivity in general

Counterexample: Two Bonds, Step by Step

- Bonds A and B, independent; each loses 100 with probability $p = 0.9\%$ (default) and 0 otherwise
- **One bond:** $P(X \leq -100) = p = 0.9\%$, not above 1%, so $q_{1\%} = 0$ and $\text{VaR}_{1\%}(A) = \text{VaR}_{1\%}(B) = 0$
 - ▶ $\text{ES}_{1\%}(A) = (100 \cdot 0.009 + 0 \cdot 0.001)/0.01 = 90$
- **Portfolio A + B:** $P(X \leq -100) = 1 - (1 - p)^2 = 1.79\% > 1\%$, $P(X \leq -200) = p^2 = 0.0081\%$, so $q_{1\%} = -100$
 - ▶ $\text{VaR}_{1\%}(A + B) = 100 > 0 + 0 = \text{VaR}_{1\%}(A) + \text{VaR}_{1\%}(B)$
 - ▶ General formula for a discrete loss $L = -X$: $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L \mathbf{1}\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha (\alpha - P(L > \text{VaR}_\alpha)))$
- $\text{ES}_{1\%}(A + B) = 100.81 \leq 180 = \text{ES}_{1\%}(A) + \text{ES}_{1\%}(B)$

Subadditivity: VaR Fails, ES Holds



□ Blue: sum of the two stand-alone measures; red: the measure of the combined portfolio

What the Counterexample Teaches

- VaR ignores everything beyond the quantile: each bond alone “hides” its default in the last 1%
 - ▶ Together, the chance of at least one default is $1.79\% > 1\%$ and VaR jumps to 100
- A VaR-based limit would reward **concentration**: holding only A looks riskless
- ES sees the size of the tail and is coherent [Acerbi & Tasche \(2002\)](#)
 - ▶ Also convex, so it can be minimised by linear programming [Rockafellar & Uryasev \(2002\)](#)
- Credit portfolios, options and other skewed pay-offs are where VaR fails in practice

ES as an Optimisation Problem: Rockafellar–Uryasev

- **Theorem Rockafellar & Uryasev (2000):** for any integrable loss L
 - ▶ $\text{ES}_\alpha(L) = \min_{v \in \mathbb{R}} \left\{ v + \frac{1}{\alpha} \mathbb{E}[(L - v)_+] \right\}$, with $\text{VaR}_\alpha(L)$ a minimiser
- **Proof:** $F(v) = v + \mathbb{E}[(L - v)_+]/\alpha$ is convex, with right derivative $1 - P(L > v)/\alpha$
 - ▶ It is ≥ 0 exactly when $P(L > v) \leq \alpha$, i.e. $v \geq \text{VaR}_\alpha$; $F(\text{VaR}_\alpha)$ is the discrete ES formula of the two-bond slide
- **Corollary:** $(w, v) \mapsto v + \mathbb{E}[(-w^\top R - v)_+]/\alpha$ is jointly convex; minimising over v keeps convexity
 - ▶ ES is convex in w ; with positive homogeneity this **proves** subadditivity
- S&P 500 losses, $\alpha = 2.5\%$: $\arg \min F = 2.51\%$ (historical VaR), $\min F = 3.76\%$ (historical ES)

Minimum-ES Portfolio: a Linear Program

- With N historical scenarios R_s , the Rockafellar–Uryasev objective becomes a **linear program** Rockafellar & Uryasev (2000)
 - ▶ $\min_{w, v, u} v + \frac{1}{\alpha N} \sum_s u_s$ subject to $u_s \geq -w^\top R_s - v$, $u_s \geq 0$, $\sum_i w_i = 1$, $w \geq 0$
- SPY, TLT, GLD, Bitcoin, 3 017 common days, $\alpha = 2.5\%$: weights 31% / 44% / 25% / 0%
 - ▶ $ES_{2.5\%} = 1.67\%$, against 2.15% for the 40/30/20/10 portfolio of Section 6
- Long-only minimum variance: 35% / 43% / 22% / 0%, $ES_{2.5\%} = 1.67\%$: for these assets the two criteria almost agree
- They part ways for options and credit, where the tail is not a function of the variance
- The objective is piecewise linear: several portfolios can attain the same minimum, so report the solution together with its tail scenarios

When Is VaR Subadditive After All?

- **Jointly elliptical** losses (multivariate Normal, multivariate Student- t) and $\alpha < 0.5$: VaR is subadditive [McNeil, Frey & Embrechts \(2015\)](#)
 - ▶ There VaR and ES are both $\mu_p + \sigma_p \times \text{constant}$, with $\sigma_p \leq \sum_i \sigma_i$; Normal or t marginals alone are not enough
- **Heavy but not too heavy tails**: for multivariate regularly varying losses with tail index above 1, VaR is subadditive far enough in the tail (asymptotically, as $\alpha \rightarrow 0$) [Danielsson et al. \(2013\)](#)
 - ▶ Estimated tail indices of equity and FX returns are in that range (between 4.8 and 6.5 in Section 7), but the result is asymptotic, not a guarantee at a given α
- It fails for discrete jumps (defaults), extremely heavy tails and some option pay-offs
- VaR vs ES in practice: ES needs more data for the same precision [Yamai & Yoshida \(2005\)](#)

Coherence versus Robustness

- **Qualitative robustness:** a small change in the data distribution changes the law of the estimator only a little
Cont, Deguest & Scandolo (2010)
 - ▶ Historical VaR is qualitatively robust; historical ES is **not**: its influence function $VaR_\alpha + (x - VaR_\alpha)_+/\alpha - ES_\alpha$ is unbounded in x
- S&P 500, last 500 days: drop the single largest loss (6.16%, 2025-04-04)
 - ▶ $ES_{2.5\%}$: 2.94% $\rightarrow$ 2.60%; $VaR_{1\%}$: 2.73% $\rightarrow$ 2.68%
- A trade-off, not a free lunch: subadditivity needs the tail beyond the quantile, which is exactly what makes ES sensitive to single observations
- FRTB chose coherence for capital (ES 2.5%) and kept robust, elicitable VaR for backtesting

VaR Aggregation under Dependence Uncertainty

- Margins $F_1, \dots, F_d$ known, copula unknown (Chapter 6): $\text{VaR}_\alpha(\sum_i L_i)$ lies in an interval Embrechts, Puccetti & Rüschendorf (2013)
 - ▶ The comonotone value $\sum_i \text{VaR}_\alpha(L_i)$ is **not** the upper end: VaR is not subadditive
- **Rearrangement algorithm**: discretise the tail beyond each $\text{VaR}_\alpha(L_i)$ in N points (one column per asset)
 - ▶ Order each column opposite to the sum of the others until the row sums are flat; worst VaR $\approx$ smallest row sum
- 40/30/20/10 portfolio of Section 6, daily $\text{VaR}_{1\%}$, empirical margins, $N = 2\,000$
 - ▶ Historical dependence 1.95%; comonotone 3.63%; worst case 4.76% ($1.31 \times$ comonotone)
- Capital based on VaR carries dependence model risk of the same order as the diversification benefit

Preview: Elicitability (Chapter 8)

- A risk measure is **elicitable** if it is the minimiser of an expected scoring function
 - ▶ VaR: the quantile $q_\alpha(X)$ minimises the pinball loss $\mathbb{E}[(1\{X \leq v\} - \alpha)(v - X)]$
- ES alone is **not** elicitable [Gneiting \(2011\)](#): no score ranks ES forecasts correctly on its own
- But the pair (VaR, ES) is jointly elicitable [Fissler & Ziegel \(2016\)](#)
- Consequence: comparing ES forecasts is harder than comparing VaR forecasts
 - ▶ ES precision is governed by the tail sample $n\alpha$, not by n [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
 - ▶ S&P 500, 26 years: only 168 days in the tail of ES 2.5%

Entropy and Market Risk

- **Differential entropy** of a density f : $H(L) = - \int f(x) \ln f(x) dx$, a measure of dispersion without moments
 - ▶ For a location–scale family $L = \mu + \sigma Z$: $H(L) = \ln \sigma + H(Z)$, so $\text{VaR}_\alpha = \mu + e^{H(L)-H(Z)} q_{1-\alpha}(Z)$
- [Pele, Lazar & Dufour \(2017\)](#): the entropy of intraday returns is linked to intraday VaR and ES and helps forecast daily VaR (EUR/JPY)
- [Pele & Mazurencu-Marinescu-Pele \(2019\)](#): high-frequency entropy gives the best daily VaR forecasts for Bitcoin among the models compared
- [Pele, Lazar & Mazurencu-Marinescu-Pele \(2019\)](#): a nonparametric tail-entropy measure as an alternative estimator of ES

Recap: Coherence

- Coherence = monotonicity, translation invariance, positive homogeneity and subadditivity
- VaR can violate subadditivity (the two-bond example); ES never does: Rockafellar–Uryasev makes it a convex minimum
- Coherence costs robustness, and VaR aggregation depends on the unknown copula
- For elliptical and moderately heavy-tailed returns VaR is subadditive in practice
- VaR is elicitable; ES alone is not: this shapes backtesting in Chapter 8

Five Families of Methods

Method	Assumption	Weak point
Historical simulation	The past n days are a sample from tomorrow's distribution	Slow reaction; no loss beyond the worst seen
Parametric (Normal, t , Cornish–Fisher)	A distribution family with estimated parameters	Wrong family = wrong tail
Filtered historical simulation	GARCH volatility $\times$ empirical standardised residuals	Depends on the volatility model
Monte Carlo	Simulated paths from a fitted model	Model risk; computing time
Extreme value theory	Only the tail follows a limit law	Threshold choice; few data

- Every method gives VaR and ES from the same fitted distribution: ES_α is the average of VaR_u , $u < \alpha$

Historical Simulation

- Take the last n daily losses $L_{t-n+1}, \dots, L_t$ and sort them: $L_{(1)} \leq \dots \leq L_{(n)}$
 - ▶ $\widehat{\text{VaR}}_\alpha =$ empirical $(1 - \alpha)$ -quantile, interpolated linearly at position $1 + (n - 1)(1 - \alpha)$ of the sorted losses (the convention used throughout)
 - ▶ $\widehat{\text{ES}}_\alpha =$ mean of the k_α losses $\geq \widehat{\text{VaR}}_\alpha$, $k_\alpha \approx n\alpha$; the exact plug-in of the integral gives the boundary loss the weight $n\alpha - \lfloor n\alpha \rfloor$, so the two coincide only when $n\alpha$ is an integer (distinct losses)
- No distributional assumption: heavy tails and skewness are kept automatically
- Choice of window n : 250 days reacts faster, 1000 days is more precise
 - ▶ With $n = 500$: 5 losses lie beyond VaR 1%; ES 2.5% averages $k = 13$ losses ($n\alpha = 12.5$)
- **Ghost effect**: a crash day enters the window, stays for n days, then drops out and VaR falls abruptly

How Precise Are Historical VaR and ES? Asymptotic Theory

- ▣ **VaR**: for i.i.d. losses with density f continuous and positive at VaR_α ,

$$\sqrt{n}(\widehat{\text{VaR}}_\alpha - \text{VaR}_\alpha) \rightarrow N\left(0, \frac{\alpha(1-\alpha)}{f(\text{VaR}_\alpha)^2}\right)$$
 - ▶ The density is small in the tail: $\sigma = 1.2\%$, $n = 500$, $\text{VaR}_{1\%}$ has SE 0.20 percentage points under the Normal distribution and 0.43 under a t_4 with the same σ
- ▣ **ES** Chen (2008): influence function $\text{IF}(x) = \text{VaR}_\alpha + (x - \text{VaR}_\alpha)_+/\alpha - \text{ES}_\alpha$
 - ▶ No term from estimating VaR: $\partial F/\partial v = 0$ at $v = \text{VaR}_\alpha$ (Rockafellar–Uryasev); if $\mathbb{E}[(L - \text{VaR}_\alpha)_+^2] < \infty$, the variance of the limit of $\sqrt{n}(\widehat{\text{ES}}_\alpha - \text{ES}_\alpha)$ is $\text{avar} = \text{Var}((L - \text{VaR}_\alpha)_+)/\alpha^2$
 - ▶ Pareto-type tail with index $a > 2$, $\alpha \rightarrow 0$: $\text{Var}((L - \text{VaR}_\alpha)_+) \approx \frac{2\alpha \text{VaR}_\alpha^2}{(a-1)(a-2)}$, so $\text{SE} \propto \text{VaR}_\alpha/\sqrt{n\alpha}$: the **relative** error falls like $1/\sqrt{n\alpha}$, and the tail count $n\alpha$ governs precision Pele & Mazurencu-Marinescu-Pele (2026)
- ▣ Volatility clustering (stationary, weakly dependent losses): replace the variances by long-run (HAC) variances of $1\{L_t > \text{VaR}_\alpha\}$ and $(L_t - \text{VaR}_\alpha)_+$ Newey & West (1987), or use a block bootstrap

Precision on the S&P 500

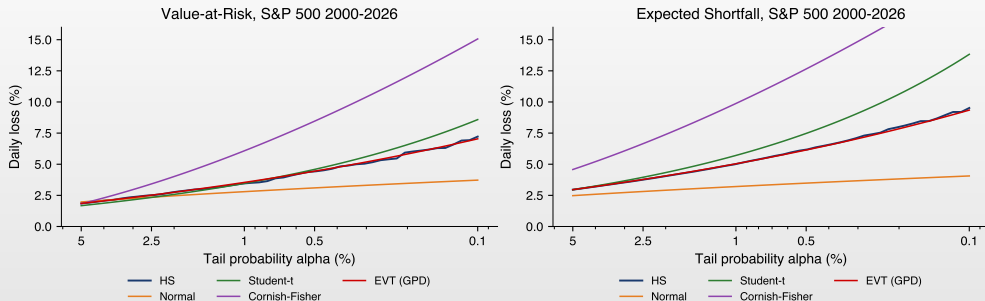
	n	$\widehat{\text{VaR}}_{1\%}$	SE i.i.d.	SE HAC	$\widehat{\text{ES}}_{2.5\%}$	SE i.i.d.	SE HAC
2000–2026	6 717	3.45	0.13	0.19	3.76	0.15	0.26
Last 500 days	500	2.73	0.36	0.44	2.94	0.48	0.65

- Density $f(\text{VaR})$ by a Gaussian kernel (Silverman bandwidth); HAC: Bartlett kernel, lag $\lfloor 4(n/100)^{2/9} \rfloor$ (10 and 5) [Newey & West \(1994\)](#)
- Full sample, width of the 95% confidence interval for ES: 0.60 (i.i.d.) and 1.00 (HAC)
 - Seminar 7, B6: i.i.d. bootstrap 0.62, block bootstrap 1.16: the asymptotic widths are close to both
- Clustering inflates the SE by 50–70% on the full sample; with 500 days ES 2.5% averages 13 losses ($n\alpha = 12.5$) and its HAC 95% interval is ± 1.27 percentage points

The Cornish–Fisher Expansion

- Adjust the Normal quantile for **skewness** $S = \mathbb{E}[(X - \mu)^3]/\sigma^3$ and **excess kurtosis** $K = \mathbb{E}[(X - \mu)^4]/\sigma^4 - 3$
Cornish & Fisher (1938)
 - ▶ $\tilde{z}_\alpha = z_\alpha + \frac{(z_\alpha^2 - 1)S}{6} + \frac{(z_\alpha^3 - 3z_\alpha)K}{24} - \frac{(2z_\alpha^3 - 5z_\alpha)S^2}{36}, \quad \text{VaR}_\alpha = -(\mu + \sigma\tilde{z}_\alpha), \text{ with } z_\alpha < 0$
- ES: average of $-(\mu + \sigma\tilde{z}_u)$ over $u \in (0, \alpha)$, numerically
- An expansion around the Normal distribution: accurate only for **small** S and K
- S&P 500 returns: $S = -0.35$, $K = 10.7$: far outside that range
 - ▶ $\text{VaR}_{1\%}^{CF} = 6.08\%$ versus historical 3.45%: the kurtosis term overshoots

Five Methods Across Tail Probabilities



□ S&P 500 daily losses 2000–2026; tail probabilities α from 5% down to 0.1%

Reading the Level Curves

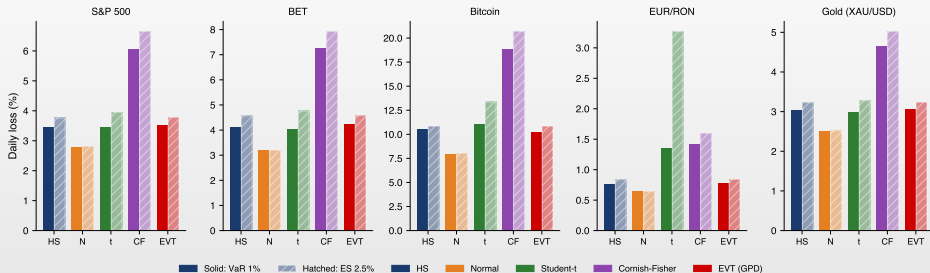
- VaR 0.1% (one day in four years): historical 7.22%, extreme-value 7.04%
 - ▶ Normal distribution: 3.72%, about half; Student- t : 8.58%; Cornish–Fisher: 15.06%
- The gap between methods **grows** as α falls: the further into the tail, the more the model matters
- ES curves spread out more than VaR curves: ES depends on the whole tail shape
- Extreme-value and historical estimates agree down to $\alpha = 0.5\%$; further into the tail historical simulation becomes sparse, and EVT extrapolates with a fitted tail model (the Normal and t fits extrapolate too, but from the whole distribution)

VaR and ES by Method and Asset

	VaR _{1%} (%)					ES _{2.5%} (%)				
	Hist.	Normal	<i>t</i>	CF	EVT	Hist.	Normal	<i>t</i>	CF	EVT
S&P 500	3.45	2.80	3.46	6.08	3.54	3.76	2.81	3.95	6.65	3.78
BET	4.13	3.19	4.06	7.25	4.22	4.57	3.21	4.76	7.93	4.58
Bitcoin	10.51	7.99	11.06	18.89	10.23	10.84	8.03	13.37	20.69	10.84
EUR/RON	0.76	0.64	1.36	1.43	0.78	0.84	0.65	3.27	1.60	0.84
Gold	3.04	2.50	3.00	4.64	3.07	3.23	2.51	3.27	5.02	3.23

- Daily losses, full samples; CF = Cornish–Fisher, EVT = generalised Pareto tail fitted to the largest 5% of losses (Section 7)
- Bitcoin needs about three times the S&P 500 buffer; EUR/RON about a fifth

Methods Compared on Five Assets



□ Solid bars: $\text{VaR}_{1\%}$; hatched bars: $\text{ES}_{2.5\%}$; daily losses in %

What Goes Wrong, and Where

- **Normal distribution:** too low everywhere, e.g. Bitcoin $\text{VaR}_{1\%}$ 7.99% vs 10.51%
 - ▶ Its ES 2.5% is almost its VaR 1%: the Normal tail is thin
- **Cornish–Fisher:** excess kurtosis above 7 in every series makes it overshoot, e.g. BET 7.25% vs 4.13%
 - ▶ Kurtosis: S&P 500 10.7, BET 11.1, Bitcoin 12.0, EUR/RON 15.1, gold 7.7
- **Student- t :** good VaR, but EUR/RON gives $\hat{\nu} = 1.25$ (infinite variance) and an absurd ES of 3.27%
 - ▶ The peak of many near-zero reference-rate changes drives $\hat{\nu}$ down
- **Extreme value theory:** close to historical simulation for VaR 1% in all five series, with a smooth tail beyond

Filtered Historical Simulation

- Idea [Barone-Adesi, Giannopoulos & Vosper \(1999\)](#), [Hull & White \(1998\)](#): history gives the **shape** of the shocks, a GARCH model gives today's **scale**
 - ▶ $r_t = \mu_t + \sigma_t z_t$, $\mu_t = c + \phi r_{t-1}$, $\sigma_t^2 = \omega + \alpha_1 (r_{t-1} - \mu_{t-1})^2 + \beta_1 \sigma_{t-1}^2$ (Chapter 5, [Bollerslev \(1986\)](#))
- Step 1: estimate the AR(1)-GARCH(1,1) model by quasi-maximum likelihood (Normal likelihood, robust to other shock distributions)
- Step 2: standardised residuals $\hat{z}_t = (r_t - \hat{\mu}_t) / \hat{\sigma}_t$: approximately i.i.d., still heavy-tailed
- Step 3: rescale by tomorrow's volatility
 - ▶ $\text{VaR}_{\alpha, t+1} = -(\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} q_\alpha(\hat{z}))$, $\text{ES}_{\alpha, t+1} = -\hat{\mu}_{t+1} - \hat{\sigma}_{t+1} \overline{(\hat{z} \mid \hat{z} \leq q_\alpha(\hat{z}))}$

GARCH Filters and Tomorrow's Volatility

	$\hat{\phi}$	$\hat{\alpha}_1$	$\hat{\beta}_1$	$\hat{\alpha}_1 + \hat{\beta}_1$	$\hat{\sigma}$ (sample)	$\hat{\sigma}_{t+1}$	$-q_1\%(\hat{z})$
S&P 500	-0.053	0.120	0.860	0.981	1.21	0.73	2.79
BET	0.104	0.207	0.787	0.994	1.40	1.76	2.74
Bitcoin	-0.009	0.127	0.843	0.970	3.49	3.00	2.88
EUR/RON	0.169	0.200	0.780	0.980	0.28	0.11	2.74
Gold	-0.022	0.046	0.943	0.989	1.09	1.43	2.68

- $\hat{\sigma}_{t+1}$: forecast for the first trading day after 18 September 2026 (19 September for Bitcoin), in %
- Persistence $\hat{\alpha}_1 + \hat{\beta}_1$ close to 1: volatility shocks die out slowly (Chapter 5)
- S&P 500 and EUR/RON are calm today ($\hat{\sigma}_{t+1}$ below the sample value); BET and gold are turbulent
- $-q_1\%(\hat{z})$ is between 2.68 and 2.88, above $-z_1\% = 2.326$: the filtered shocks are still heavy-tailed

Conditional VaR and ES for the Next Trading Day

	VaR _{1%} (%)				ES _{1%} (%)			
	GARCH	FHS	c-EVT	Hist.	GARCH	FHS	c-EVT	Hist.
S&P 500	1.63	1.97	2.03	3.45	1.88	2.55	2.56	5.01
BET	3.80	4.53	4.63	4.13	4.39	6.16	6.20	6.28
Bitcoin	6.89	8.56	8.68	10.51	7.90	12.22	12.19	14.29
EUR/RON	0.23	0.27	0.26	0.76	0.27	0.34	0.35	1.12
Gold	3.32	3.82	3.86	3.04	3.80	4.94	4.93	4.21

- GARCH: GARCH with Normal shocks; c-EVT: conditional extreme value theory (Section 7); Hist.: unconditional, full sample
- Conditional numbers move with the market: S&P 500 FHS VaR 1.97% today vs 3.45% unconditional
- GARCH with Normal shocks is the lowest of the three conditional methods in every row: volatility alone does not create the tail

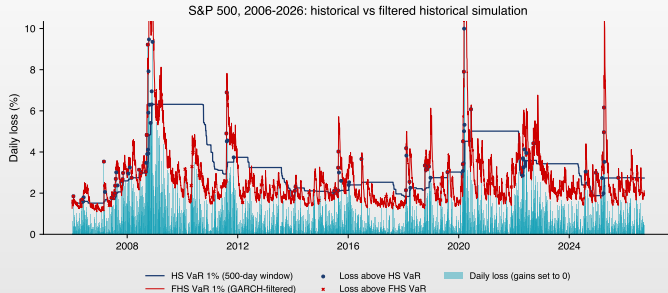
Estimation Risk in Conditional VaR

- $\text{VaR}_{\alpha,t+1} = -\mu_{t+1}(\theta) + \sigma_{t+1}(\theta) \xi_{\alpha}$, $\xi_{\alpha} = -q_{\alpha}(z)$: **both** θ (GARCH, by QML) and ξ_{α} (a residual quantile) are estimated
 - ▶ The table above reports point forecasts; a risk committee also needs their sampling error
- **Francq & Zakoïan (2015)**: estimate the **risk parameter** (θ, ξ_{α}) jointly; under strict stationarity, $\mathbb{E}[z_t^4] < \infty$ and a positive density of z at $q_{\alpha}(z)$, the conditional VaR is asymptotically Normal (QML consistency alone needs less), with a variance from $\hat{\theta}$, from $\hat{\xi}_{\alpha}$ and from their covariance
- **Residual bootstrap Christoffersen & Gonçalves (2005)**: simulate AR(1)-GARCH(1,1) paths with resampled $\hat{z}$ and re-estimate θ^* on each
 - ▶ σ_{t+1}^* from the **original** returns with θ^* ; ξ_{α}^* from the path's own residuals; percentile interval of VaR_{t+1}^*
- Heavy tails and skewness of z slow the convergence; **Spierdijk (2016)** compares interval methods for ARMA–GARCH VaR in exactly this case

A Confidence Interval for Today's FHS VaR

- ▣ S&P 500, 999 bootstrap paths: FHS $\text{VaR}_{1\%} = 1.97\%$, 90% interval $[1.90, 2.09]\%$
- ▣ The two sources: $\hat{\sigma}_{t+1} = 0.725$ $[0.708, 0.743]$; $\hat{\xi}_{1\%} = 2.79$ $[2.68, 2.88]$
 - ▶ Only the quantile varies: $[1.89, 2.03]$; only σ_{t+1} varies: $[1.92, 2.02]$: the residual quantile (about 67 tail residuals) is the larger source
- ▣ Conditional EVT (2.03%) lies inside the interval, GARCH with Normal shocks (1.63%) far below it
- ▣ Today the choice of the tail model matters more than the estimation error of a given model

Historical vs Filtered Historical Simulation: S&P 500

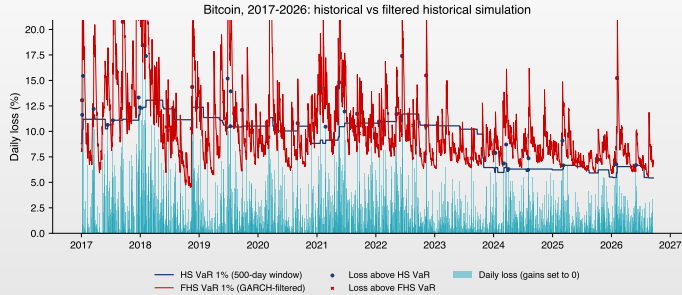


- Daily one-day-ahead $\text{VaR}_{1\%}$; GARCH parameters re-estimated each January on all earlier data, so no future information is used

Reading the S&P 500 Comparison

- ▣ Share of days with a loss above VaR 1%, 2006–2026 (5 210 days; target $\alpha = 1\%$)
 - ▶ Historical, 500 days: 1.67%; filtered: 1.54%; GARCH with Normal shocks: 2.34%
- ▣ Historical VaR moves between 1.49% and 6.32%; filtered VaR between 1.09% and 18.01% (peak on 2020-03-18)
- ▣ Historical VaR exceedances come in **clusters** (2008, 2020): the window reacts months too late
- ▣ Filtered VaR spreads exceedances over time: what a good forecast should do (formal tests in Chapter 8)

Historical vs Filtered Historical Simulation: Bitcoin



□ Same design as for the S&P 500; forecasts from January 2017, 7 days a week

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Reading the Bitcoin Comparison

- ▣ Share of days with a loss above VaR (3 548 days): historical 1.16%, filtered 0.76%, GARCH with Normal shocks 1.58%
- ▣ Filtered VaR peaked at 54.83% in March 2020, just after the crash of 2020-03-12 (log loss 46.47% in one day, a price fall of 37.17%)
- ▣ Bitcoin volatility has fallen since 2022; the 500-day historical VaR follows with a long delay
- ▣ Crypto as a distinct asset class with its own risk profile: [Pele et al. \(2023\)](#)

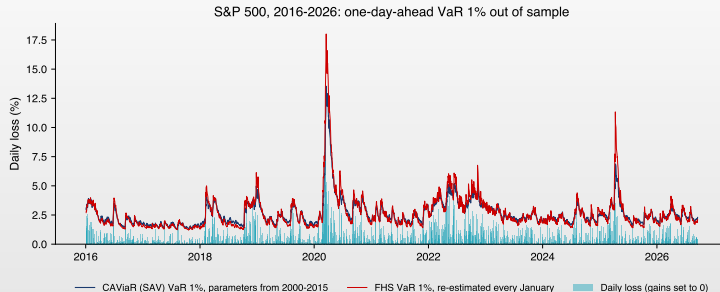
Monte Carlo Simulation

- Simulate many future scenarios from a fitted model; read VaR and ES from the simulated losses
 - ▶ Needed when the loss is a non-linear function of the risk factors (options) or the horizon is several days
- **Filtered Monte Carlo** for h days: repeat for $j = 1, \dots, M$
 - ▶ draw $z_{t+1}^{(j)}, \dots, z_{t+h}^{(j)}$ with replacement from the standardised residuals $\hat{z}$
 - ▶ update $\sigma^{(j)}$ and $r^{(j)}$ day by day with the GARCH recursion; $L^{(j)} = -\sum_{i=1}^h r_{t+i}^{(j)}$
- $\widehat{\text{VaR}}_\alpha, \widehat{\text{ES}}_\alpha =$ historical estimates on the M simulated losses; simulation error falls as $1/\sqrt{M}$
- Keeps volatility clustering, heavy tails and the mean reversion of volatility over the horizon

Modelling the Quantile Directly: CAViaR

- CAViaR (Conditional Autoregressive Value at Risk) Engle & Manganelli (2004): the quantile follows its own recursion; SAV specification
 - ▶ $\text{VaR}_t = \beta_0 + \beta_1 \text{VaR}_{t-1} + \beta_2 |r_{t-1}|, \quad q_t = -\text{VaR}_t$
- **Quantile regression** Koenker & Bassett (1978): minimise the pinball loss, no assumption on the shock distribution
 - ▶ $\hat{\beta} = \arg \min_{\beta} \sum_t (\alpha - 1\{r_t < q_t(\beta)\})(r_t - q_t(\beta))$: elicibility turned into an estimator
- Published estimation strategy: 10^4 random starts in $(0, 1)^3$, the best 10 refined by simplex and quasi-Newton steps; $\text{VaR}_1 = -\hat{q}_{\alpha}$ of the first 300 returns (their upper loss quantile), days kept in the estimation sample
- S&P 500, $\alpha = 1\%$, estimated on 2000–2015 (4 024 days): $\hat{\beta}_0 = 0.090, \hat{\beta}_1 = 0.897, \hat{\beta}_2 = 0.246$
 - ▶ Half-life of a shock to VaR: $\ln 0.5 / \ln \hat{\beta}_1 = 6.4$ days

CAViaR vs Filtered Historical Simulation, Out of Sample



- S&P 500, 2016–2026: CAViaR with parameters fixed on 2000–2015; FHS re-estimated each January on all earlier data

Reading the CAViaR Comparison

- ▣ 2016–2026 (2 693 days), share of days with a loss above VaR 1%: CAViaR 1.30% (35), FHS 1.30% (35), historical 500 days 1.41% (38)
 - ▶ Mean pinball loss (lower is better): CAViaR 0.039, FHS 0.037, historical 0.052
- ▣ COVID-19 window, 20 February – 30 April 2020 (50 days): exceedances CAViaR 6, FHS 3, historical 9
 - ▶ Peak VaR: CAViaR 13.5%, FHS 18.0%, historical 4.5%
- ▣ CAViaR reacts linearly to $|r_{t-1}|$, GARCH to r_{t-1}^2 : after a crash FHS jumps higher and has fewer exceedances
- ▣ Over the decade the two conditional models are close; whether the difference is significant is a Chapter 8 question

Joint Models for VaR and ES

- ES alone is not elicitable, but the pair (VaR, ES) is [Fissler & Ziegel \(2016\)](#): it can be **estimated** by minimising a strictly consistent joint score
 - ▶ Chapter 8 uses the same scores to **evaluate** forecasts
- [Patton, Ziegel & Chen \(2019\)](#): dynamic semiparametric (VaR, ES) models, e.g. with GAS dynamics, estimated by minimising the FZ0 loss; consistency and asymptotic normality
- [Taylor \(2019\)](#): CAViaR-type recursions for VaR and ES, estimated by an asymmetric Laplace likelihood
- [Dimitriadis & Bayer \(2019\)](#): linear regression of the quantile and of ES on covariates (realised volatility, implied volatility), with standard errors
- Common thread: the scoring function that ranks forecasts is also the objective function that estimates the model

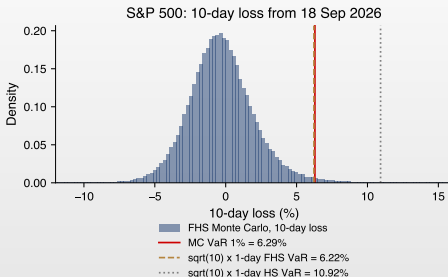
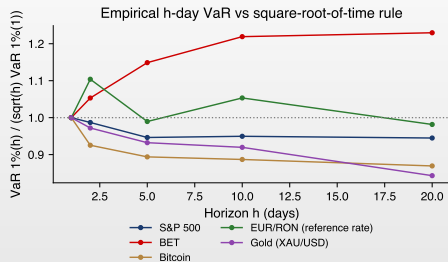
Recap: Methods

- ▣ Historical simulation: simple and model-free, but slow and blind beyond the worst observed loss; ES precision is set by $n\alpha$
- ▣ Normal distribution: underestimates the tail; Cornish–Fisher: unreliable with the kurtosis of daily returns
- ▣ Student- t : a good one-parameter tail, fragile when $\hat{\nu}$ approaches 2
- ▣ Filtered historical simulation and Monte Carlo: today's volatility + empirical shock shape; our default for daily risk, reported with a bootstrap interval
- ▣ CAViaR and joint (VaR, ES) models estimate the risk measure directly by its scoring function

The Square-Root-of-Time Rule: Assumptions and Failures

- i.i.d. $N(0, \sigma^2)$ daily losses: $\text{VaR}_\alpha^{(h)} = \sqrt{h} \text{VaR}_\alpha^{(1)}$, $\text{ES}_\alpha^{(h)} = \sqrt{h} \text{ES}_\alpha^{(1)}$; Basel 1996 allowed $\sqrt{10} = 3.162$ [Basel Committee \(1996\)](#)
 - ▶ Needs zero mean, no autocorrelation, constant volatility and Normal shocks; each fails in the data (Chapters 1, 2, 5)
- **Autocorrelation:** $\text{Var}(r_{t+1} + \dots + r_{t+h}) = h\sigma^2 \text{VR}(h)$ (variance ratio, Chapter 2)
 - ▶ $\text{VR} > 1$: $\sqrt{h}$ **underestimates** the h -day risk
- **Volatility clustering:** volatility reverts to its mean over h days; calm today $\Rightarrow \sqrt{h}$ too low, turbulent today $\Rightarrow$ too high
- **Heavy tails:** for i.i.d. shocks with tail index $a > 2$ the far-tail quantile of a sum grows like $h^{1/a} < \sqrt{h}$ [Dánielsson \(2011\)](#)
 - ▶ With jumps the rule can badly misstate multi-day risk [Dánielsson & Zigrand \(2006\)](#)

Square-Root-of-Time in the Data



- Left: empirical $\text{VaR}_{1\%}$ of overlapping h -day losses divided by $\sqrt{h}$ times the one-day value; right: filtered Monte Carlo, 200 000 paths

Reading the Horizon Chart

- Ratio at $h = 10$: BET 1.22 (lag-1 autocorrelation 0.109), EUR/RON 1.05 (0.170)
 - ▶ S&P 500 0.95 (-0.099), gold 0.92, Bitcoin 0.89: heavy tails and reversal make $\sqrt{h}$ conservative
- S&P 500 today: $\hat{\sigma}_{t+1} = 0.73\%$ vs 1.21% in the sample
 - ▶ 10-day $\text{VaR}_{1\%}$: filtered Monte Carlo 6.29% ; $\sqrt{10} \times$ one-day filtered 6.22% ; $\sqrt{10} \times$ one-day historical 10.92%
- The mean reversion of volatility and the heavy tails of the shocks roughly offset each other here
- The big difference is conditional vs unconditional, not $\sqrt{h}$ vs simulation

Recap: Horizon

- ▣ $\sqrt{h}$ is exact only for i.i.d. Normal losses with zero mean
- ▣ Positive autocorrelation (BET) makes it too low; heavy tails and reversal (S&P 500, Bitcoin) too high
- ▣ Simulation from a filtered model handles all effects at once
- ▣ Direct h -day historical estimates use overlapping sums: fewer independent observations

Portfolio VaR and Euler Allocation

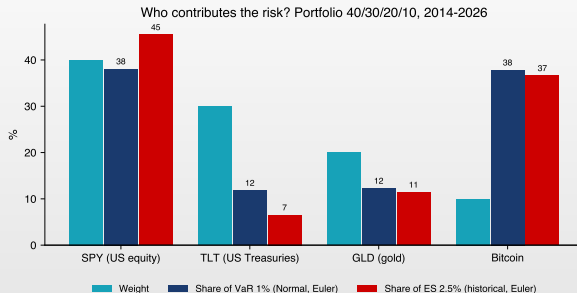
- Weights w , simple returns R with covariance Σ (simple returns add across assets); $L_p = -w^\top R$
 - ▶ Normal, mean ignored: $\text{VaR}_\alpha(L_p) = z_{1-\alpha}\sigma_p$, $\sigma_p = \sqrt{w^\top \Sigma w}$; undiversified VaR $\sum_i z_{1-\alpha} w_i \sigma_i$ is larger unless all correlations are 1
- **Marginal VaR** $\partial \text{VaR}_\alpha / \partial w_i = z_{1-\alpha}(\Sigma w)_i / \sigma_p$; **component VaR** $C_i = w_i \partial \text{VaR}_\alpha / \partial w_i$
 - ▶ Euler's theorem for a function homogeneous of degree 1: $\sum_i w_i \partial \rho / \partial w_i = \rho$, so $\sum_i C_i = \text{VaR}_\alpha$ [Tasche \(2008\)](#)
- ES, no distributional assumption (continuous losses): $C_i^{ES} = \mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_\alpha(L_p)]$, $\sum_i C_i^{ES} = \text{ES}_\alpha(L_p)$
 - ▶ Historical estimate: the average over the tail days; a negative C_i means the asset **hedges** the portfolio

The Portfolio: SPY, TLT, GLD and Bitcoin

	Weight	σ_i (%)	corr. SPY	corr. TLT	corr. GLD
SPY	40%	1.10	1.00		
TLT	30%	0.93	-0.17	1.00	
GLD	20%	1.03	0.07	0.25	1.00
Bitcoin	10%	4.14	0.23	-0.02	0.09

- Daily simple returns on 3017 common trading days, September 2014 – September 2026; daily rebalancing
- Portfolio $\sigma_p = 0.76\%$; $\text{VaR}_{1\%}$ Normal = 1.77%, undiversified 3.11%: diversification removes 43%
- Historical $\text{ES}_{2.5\%} = 2.15\%$ vs undiversified 3.87% (44% less)

Who Contributes the Risk?



□ Teal: weights; blue: shares of Normal $\text{VaR}_{1\%}$; red: shares of historical $\text{ES}_{2.5\%}$ (Euler allocation)

Reading the Risk Contributions

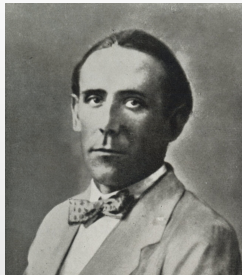
	Marginal VaR	C_i	share	C_i^{ES}	share
SPY	1.68	0.67	38.1%	0.98	45.4%
TLT	0.70	0.21	11.8%	0.14	6.5%
GLD	1.08	0.22	12.3%	0.25	11.4%
Bitcoin	6.68	0.67	37.8%	0.79	36.6%
Total		1.77	100%	2.15	100%

- Bitcoin: 10% of the money, 37.8% of VaR and 36.6% of ES: its volatility is almost four times that of SPY
- TLT: 30% of the money, only 6.5% of ES: on the portfolio's worst days TLT lost only 0.47% on average
- SPY carries more of the ES than of the Normal VaR (45.4% vs 38.1%): its losses concentrate on the portfolio's worst days more than covariances imply; calling this tail dependence needs a separate analysis (Chapter 6)

Recap: Portfolio Risk

- Under joint Normality, portfolio VaR is below the sum of stand-alone VaRs whenever correlations are below 1; outside elliptical models it can exceed it (two-bond example)
- Euler allocation splits VaR or ES into contributions that add up exactly
- Risk shares, not money weights, show where the risk is: small positions can dominate
- Historical ES contributions reflect co-movement on the worst days, which covariance-based contributions need not capture (tail dependence: Chapter 6)

Emil Gumbel and the Statistics of Extremes



Emil Julius Gumbel (1891–1966)

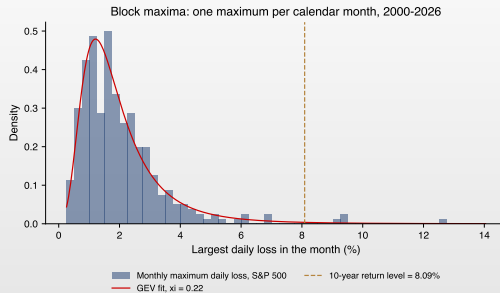
Photo: unknown author (c. 1931); public domain;
Wikimedia Commons

- Fisher & Tippett (1928): the maximum of many i.i.d. variables, suitably rescaled, can have only three limit forms
- Gnedenko (1943) proved the result rigorously
- Gumbel (1958) wrote the first textbook and applied it to floods, droughts and life lengths
 - ▶ The Gumbel distribution is one of the three limit forms
- In finance: model **only the tail**, where the risk is Embrechts, Klüppelberg & Mikosch (1997), Longin (2000)

Block Maxima and the GEV Distribution

- Split the sample into blocks of n days; keep each block maximum $M_n = \max(L_1, \dots, L_n)$
 - ▶ Fisher–Tippett–Gnedenko theorem: if $(M_n - b_n)/a_n$ has a non-degenerate limit, the limit is GEV
- $H_\xi(x) = \exp \left\{ - \left(1 + \xi \frac{x - \mu}{\sigma} \right)^{-1/\xi} \right\}$, $1 + \xi(x - \mu)/\sigma > 0$ ($\xi = 0$: $\exp\{-e^{-(x - \mu)/\sigma}\}$)
- The **shape** ξ decides the tail
 - ▶ $\xi > 0$: Fréchet, power-law tail with tail index $1/\xi$ (financial returns)
 - ▶ $\xi = 0$: Gumbel domain, exponential-type tails (e.g. Normal, lognormal); $\xi < 0$: Weibull, bounded tail
- **Return level** for k blocks: the level exceeded on average once every k blocks, $H_\xi^{-1}(1 - 1/k)$

Monthly Maxima of S&P 500 Losses



- 321 calendar months, 2000–2026; GEV fitted by maximum likelihood

Reading the Block Maxima

- $\hat{\xi} = 0.22$ (Fréchet), $\hat{\mu} = 1.37$, $\hat{\sigma} = 0.79$: a power-law tail with index about $1/\hat{\xi}$
 - ▶ 10-year return level: 8.09%; 30-year: 10.95%; the largest monthly maximum was 12.77%
- From monthly maxima to a daily VaR: $P(M_n \leq x) \approx P(L \leq x)^n$, so $\text{VaR}_\alpha \approx H^{-1}((1 - \alpha)^n)$ with $n = 20.9$
 - ▶ $\text{VaR}_{1\%} \approx H^{-1}(0.810) = 2.83\%$ vs historical 3.45%; $\text{VaR}_{0.1\%} \approx 6.19\%$
- The daily approximation assumes independent days; volatility clustering breaks it (extremal index, below), and it is too low here
- Block maxima throw away data (one value per month): the next method uses all large losses

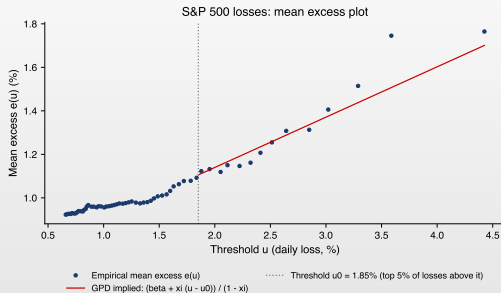
Peaks over Threshold and the GPD

- Fix a high threshold u ; model the **excesses** $Y = L - u$ given $L > u$
 - ▶ Pickands–Balkema–de Haan theorem [Pickands \(1975\)](#), [Balkema & de Haan \(1974\)](#): for a wide class of distributions, as u approaches the right end-point of the distribution the excesses follow a generalised Pareto distribution
- $G_{\xi, \beta}(y) = 1 - (1 + \xi y / \beta)^{-1/\xi}$ ($\xi \neq 0$), $1 - e^{-y/\beta}$ ($\xi = 0$); $y \geq 0$, and $y < -\beta/\xi$ if $\xi < 0$; scale $\beta > 0$; the same ξ as the GEV
- Tail estimator with N_u exceedances among n observations [McNeil, Frey & Embrechts \(2015\)](#)
 - ▶ $\widehat{\text{VaR}}_\alpha = u + \frac{\hat{\beta}}{\hat{\xi}} \left[\left(\frac{n}{N_u} \alpha \right)^{-\hat{\xi}} - 1 \right], \quad \widehat{\text{ES}}_\alpha = \frac{\widehat{\text{VaR}}_\alpha}{1 - \hat{\xi}} + \frac{\hat{\beta} - \hat{\xi}u}{1 - \hat{\xi}} \quad (\alpha < N_u/n, \hat{\xi} \neq 0; \text{ES needs } \hat{\xi} < 1)$
- Bias–variance trade-off: a low u violates the limit theorem, a high u leaves few exceedances

Choosing the Threshold: the Mean Excess Function

- **Mean excess function:** $e(u) = \mathbb{E}[L - u \mid L > u]$
 - ▶ Empirical version: average of the excesses above u , plotted against u
- For a GPD tail $e(u)$ is **linear** in u
 - ▶ $e(u) = \frac{\beta + \xi(u - u_0)}{1 - \xi}$ for $u \geq u_0$, finite only if $\xi < 1$; slope $\xi/(1 - \xi)$: upward for heavy tails, flat for exponential tails
- Rule of thumb: choose u where the plot becomes roughly linear; check the stability of $\hat{\xi}$ across thresholds
- The Hill estimator of Chapter 1 [Hill \(1975\)](#) estimates the same tail index $1/\xi$

Mean Excess Plot: S&P 500

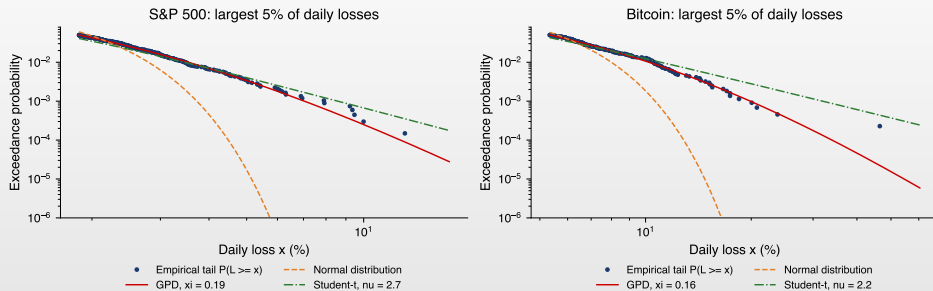


- Thresholds leaving from 20% down to 0.5% of daily losses above them; red line: GPD fitted to the largest 5%

Reading the Mean Excess Plot

- ▣ The empirical mean excess rises with u : a heavy (Fréchet-type) tail, not the flat line of an exponential tail
- ▣ At $u_0 = 1.85\%$ (top 5% of losses above it) the mean excess is 1.09%; the GPD slope $\hat{\xi}/(1 - \hat{\xi}) = 0.23$ follows the points
- ▣ Beyond VaR 1% the points scatter: each is an average of very few losses
- ▣ Keeping the top 5% leaves $N_u = 336$ exceedances: enough for a stable fit

GPD Tails: S&P 500 and Bitcoin



□ Log-log scale: probability of a daily loss above x , for the largest 5% of losses

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Reading the GPD Fits

	u (%)	N_u	$\hat{\xi}$	$SE(\hat{\xi})$	$1/\hat{\xi}$	$VaR_{1\%}$	$ES_{1\%}$
S&P 500	1.85	336	0.19	0.065	5.3	3.54	5.04
BET	1.93	334	0.21	0.066	4.8	4.22	6.34
Bitcoin	5.33	220	0.16	0.078	6.1	10.23	14.36
EUR/RON	0.38	268	0.18	0.072	5.5	0.78	1.14
Gold	1.68	348	0.15	0.062	6.5	3.07	4.22

- $\hat{\xi}$ between 0.15 and 0.21 for all five series, each more than two i.i.d. standard errors above 0: heavy tails everywhere (clustering: next slides)
- Tail index $1/\hat{\xi}$ between 4.8 and 6.5: the fitted upper tails have finite fourth moments, but every 95% interval for ξ reaches above 0.25, so this is not established (compare Chapter 1)
- On the chart the Normal tail falls far below the data; the Student- t tail is close to the GPD
- $SE(\hat{\xi}) = (1 + \hat{\xi})/\sqrt{N_u}$ assumes independent exceedances; clustering makes it too small (next slides)

Extremes of Dependent Series: the Extremal Index

- Stationary losses with clustering: if $nP(L > u_n) \rightarrow \tau$, then $P(M_n \leq u_n) \rightarrow e^{-\theta\tau}$, $\theta \in (0, 1]$ [Embrechts, Klüppelberg & Mikosch \(1997\)](#)
 - ▶ θ = extremal index, $1/\theta$ = mean cluster size; i.i.d. data: $\theta = 1$
 - ▶ Block maxima: $P(M_n \leq x) \approx F(x)^{n\theta}$: the n days act like $n\theta$ independent days
- **Intervals estimator** [Ferro & Segers \(2003\)](#) from the N exceedance times of u and the gaps T_i between them
 - ▶ $\hat{\theta} = \min \left\{ 1, \frac{2(\sum_i (T_i - 1))^2}{(N - 1) \sum_i (T_i - 1)(T_i - 2)} \right\}$ if $\max T_i > 2$ (else with T_i, T_i^2)
- **Declustering** [Ferro & Segers \(2003\)](#): target $C = \lfloor \hat{\theta}N \rfloor + 1$ clusters, separated by the $C - 1$ largest gaps; fit the GPD to the cluster maxima
 - ▶ Ties: if the $(C - 1)$ -th and C -th largest gaps are equal, C is reduced until they differ, so the realised number of clusters can be below the target

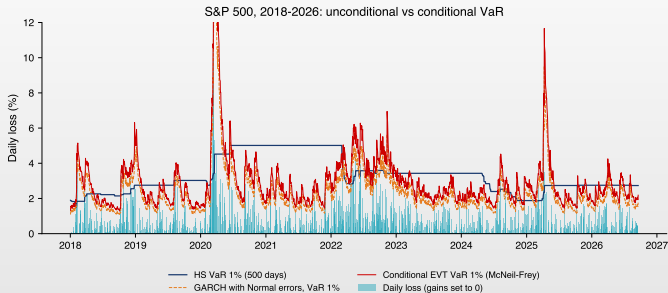
Extremal Index of S&P 500 Losses

- GPD threshold used above ($u = 1.85\%$, $N = 336$): $\hat{\theta} = 0.19$, target $C = 63$, 62 clusters after the tie rule, 5.4 exceedances each on average
 - ▶ At the historical VaR 1% ($u = 3.45\%$, $N = 68$): $\hat{\theta} = 0.35$
- GPD on cluster maxima: $\hat{\xi} = 0.152$ (SE 0.146) vs 0.188 (SE 0.065) on all exceedances: the point estimate stays positive but is not distinguishable from 0 (its 95% interval includes 0); the i.i.d. SE was less than half the honest one
- Monthly maxima with $\hat{\theta} = 0.35$: $\text{VaR}_{1\%} \approx H^{-1}(0.929) = 4.14\%$ instead of 2.83% (historical 3.45%); with $\hat{\theta} = 0.19$: 5.09%
- The correction moves the answer by more than the error it fixes: the block-maxima route to a daily VaR is fragile
- Standardised GARCH residuals: $\hat{\theta} = 0.99$: filtering removes the clustering, which is why conditional EVT fits the GPD after the filter

Conditional EVT (McNeil and Frey)

- Unconditional EVT ignores volatility clustering; [McNeil & Frey \(2000\)](#) combine GARCH filtering with a GPD tail
 - ▶ Step 1: fit AR(1)-GARCH(1,1) by quasi-maximum likelihood; standardised residuals $\hat{z}_t$
 - ▶ Step 2: fit a GPD to the largest 10% of $-\hat{z}_t$ and compute its VaR $\hat{z}_\alpha^{GPD}$ and ES at level α
 - ▶ Step 3: $\text{VaR}_{\alpha, t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{z}_\alpha^{GPD}$, with $\hat{z}_\alpha^{GPD}$ the GPD VaR of $-\hat{z}$; ES in the same way
- Same as filtered historical simulation, but with a smooth parametric tail beyond the data
- Matters most far in the tail (VaR 0.5%, VaR 0.1%), where only a handful of residuals exist

Conditional EVT on the S&P 500



□ One-day-ahead $\text{VaR}_{1\%}$, 2018–2026; parameters re-estimated each January on all earlier data

Reading the Conditional EVT Chart

- Share of days with a loss above VaR 1%, 2018–2026 (2 190 days; target $\alpha = 1\%$)
 - ▶ Conditional EVT 1.23%; GARCH with Normal shocks 2.42%; historical, 500 days 1.60%
- The GARCH–Normal line has the right **timing** but the wrong **level**: its exceedance rate is more than twice the target
- Conditional EVT keeps the timing and fixes the level: the tail of the shocks, not only their scale
- Whether 1.2% is statistically different from 1% is a question for Chapter 8

Recap: Extreme Value Theory

- ▣ Two limit theorems: block maxima $\rightarrow$ GEV, excesses over a high threshold $\rightarrow$ GPD; both governed by ξ
- ▣ All five series: $\hat{\xi} \approx 0.15\text{--}0.22$, a Fréchet-type heavy tail
- ▣ Peaks over threshold uses the data better than block maxima; the mean excess plot guides the threshold
- ▣ Clustering ($\hat{\theta} < 1$) shrinks the effective sample: decluster or filter before quoting a standard error
- ▣ Conditional EVT = GARCH scale + GPD tail: the most complete one-day model in this chapter

Liquidity-Adjusted VaR

- VaR assumes positions can be sold at the mid price; in reality we pay half the bid–ask spread
 - ▶ **Relative spread** $S = (\text{ask} - \text{bid})/\text{mid}$; it widens exactly when markets fall
- [Bangia, Diebold, Schuermann & Stroughair \(2001\)](#): $\text{LVaR}_\alpha = \text{VaR}_\alpha + \frac{1}{2}(\mu_S + a\sigma_S)$, with mean μ_S and standard deviation σ_S of the spread and a multiplier a
- Banca Transilvania, 1 million RON, last two years: $\text{VaR}_{1\%} = 3.16\%$; with assumed $\mu_S = 0.5\%$, $\sigma_S = 0.3\%$, $a = 3$: $\text{LVaR} = 3.86\%$ (+22%)
- Large positions also move the price (endogenous liquidity): the regulator handles this with longer horizons

The Basel Committee



The BIS tower in Basel, seat of the Basel Committee on Banking Supervision

Photo: Michal Pleskiewicz (2018); CC BY-SA 4.0;
Wikimedia Commons

- The Basel Committee on Banking Supervision sets global capital standards; national supervisors (in the EU, via the Capital Requirements Regulation) turn them into law
- **1996**: market risk enters the capital rules; banks may use internal VaR models [Basel Committee \(1996\)](#)
- **2019**: the Fundamental Review of the Trading Book standard replaces VaR by ES in the internal-models approach [Basel Committee \(2019\)](#); each jurisdiction sets its own application date
- EU: [Delegated Regulation \(EU\) 2025/1496](#) postponed the FRTB own-funds requirements to 1 January 2027; until then banks apply the earlier VaR-based framework

Basel 1996: VaR 1% over 10 Days

- ▣ VaR 1% (one-tailed), computed daily, minimum holding period of 10 trading days [Basel Committee \(1996\)](#)
- ▣ One-day VaR may be scaled to 10 days by the square root of time
- ▣ At least one year of historical data
- ▣ Capital: the larger of yesterday's VaR and the 60-day average VaR times a multiplication factor of at least 3
 - ▶ A “plus” of 0 to 1 is added when backtesting finds too many exceptions (Chapter 8)
- ▣ Critique after 2008: VaR ignores the size of tail losses, and one-year windows forget crises

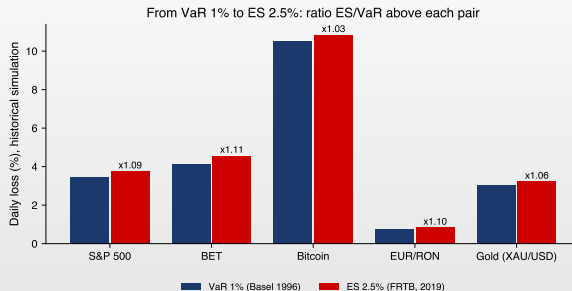
FRTB (2019): Expected Shortfall 2.5%

- ▣ ES 2.5% (one-tailed), computed daily for every trading desk using internal models [Basel Committee \(2019\)](#)
- ▣ Base horizon of 10 days, scaled up by **liquidity horizons** per risk factor
 - ▶ Large-cap equity 10 days, small-cap equity 20 days, specified currency pairs 10 days, credit spreads up to 120 days
- ▣ Calibrated to the most severe 12-month period of stress available
- ▣ Capital multiplier $m_c = 1.5$, increased by supervisors after poor backtesting
- ▣ Desks are still backtested on one-day VaR 2.5% and VaR 1%: under FRTB, VaR stays for backtesting and ES sets the capital

Why ES 2.5% Replaces VaR 1%

- Under the Normal distribution the two are almost equal
 - ▶ $ES_{2.5\%} = \sigma \times 2.338$ vs $VaR_{1\%} = \sigma \times 2.326$: the same capital for thin tails
- Student- t : $ES_{2.5\%}/VaR_{1\%}$ rises as ν falls (1.01 at $\nu = 30$, 1.07 at $\nu = 4$), and it exceeds 1 in all five samples of this chapter
 - ▶ Heavy tails alone do not guarantee it: $ES_{\alpha} \geq VaR_{\alpha}$ holds at the same α , not across levels
- ES is coherent, so desk-level ES limits add up safely
- Price: ES is harder to backtest (not elicitable alone) and less precise than VaR for the same sample

VaR 1% vs ES 2.5% on Five Assets



- Historical simulation, daily losses, full samples; the number above each pair is $ES_{2.5\%}/VaR_{1\%}$

Reading the Basel Comparison

- $ES_{2.5\%}/VaR_{1\%}$: S&P 500 1.09, BET 1.11, Bitcoin 1.03, EUR/RON 1.10, gold 1.06
 - ▶ Normal distribution: $2.338/2.326 \approx 1.005$
- All five ratios are above the Normal value: the switch raises capital by 3–11% for the same positions
- The increase is largest for BET and EUR/RON, whose tails beyond VaR 1% are longest relative to VaR itself
- Regulatory capital also adds the stressed calibration and liquidity horizons: see Seminar 7, Part C

Recap: Liquidity and Regulation

- ▣ Liquidity-adjusted VaR adds half the (stressed) spread to the market VaR
- ▣ Basel 1996: VaR 1% over 10 days, $\sqrt{10}$ scaling allowed, multiplier at least 3
- ▣ FRTB 2019: ES 2.5%, 10-day base horizon, liquidity horizons, stressed calibration, multiplier 1.5
- ▣ Under FRTB, VaR remains the backtesting tool and ES sets the capital (EU: from 1 January 2027)

An Open Question

- Can daily data rank competing (VaR 1%, ES 1%) forecasting models, given that ES alone is not elicitable and only about 1% of days are informative?
- The question is live
 - ▶ [Nolde & Ziegel \(2017\)](#): comparative backtesting with strictly consistent scores for (VaR, ES), instead of traditional coverage tests
 - ▶ [Patton, Ziegel & Chen \(2019\)](#): the FZ0 loss and dynamic semiparametric (VaR, ES) models; [Bayer & Dimitriadis \(2022\)](#): regression-based ES backtests
 - ▶ [Pele & Mazurencu-Marinescu-Pele \(2026\)](#): finite-sample precision of ES is governed by the $n\alpha$ tail observations
- Formal statement for models A and B, return Y_t , $v_t = -\text{VaR}_{\alpha,t}$, $e_t = -\text{ES}_{\alpha,t}$
 - ▶ $S(v, e; Y) = -\frac{1}{\alpha e} \mathbf{1}\{Y \leq v\}(v - Y) + \frac{v}{e} + \ln(-e) - 1$ (FZ0); $d_t = S_t^A - S_t^B$
 - ▶ $H_0: \mathbb{E}[d_t] = 0$; $H_1: \mathbb{E}[d_t] \neq 0$ (two-sided; $\bar{d} < 0$ favours A); DM t-statistic with HAC errors; falsified if the sign of $\bar{d}$ flips across pre-registered subperiods

The Discovery Loop, Step by Step (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List papers since 2015 on comparative backtesting of (VaR, ES) with strictly consistent scoring functions: DOI, scoring function, test, sample size and power findings."*
 - ▶ Resolve every DOI on Crossref; check the sign convention of the score in the paper
- **Hypotheses:** *"For FHS, conditional EVT and GARCH with Normal shocks, state which features of the tail each captures and what each predicts for the FZ0 differences in calm and in turbulent years."*
- **Data and code:** *"Write Python that computes the FZ0 loss for daily VaR 1% and ES 1% forecasts on the S&P 500 from data/market, with the sign convention $VaR = -q_\alpha$, and a Diebold–Mariano test with HAC errors."*
 - ▶ Replicate first: the FHS exceedance rate 1.54% over 5 210 days (Section 4)

The Discovery Loop, Step by Step (2/2)

- **Robustness:** *“Write a pre-registered plan: subperiods 2006–2015 and 2016–2026, levels 1% and 2.5%, the FZ0 and a second consistent score, HAC lags 5 and 20, a second index.”*
 - ▶ Size check: simulate from a fitted GARCH-t and score two forecasts with equal expected FZ0; the test must reject in about 5% of simulations
- **Critique:** *“Act as a referee: is my ranking driven by 2008 and 2020, by the score’s sign convention or by overlapping estimation windows? Give the test that would expose each.”*
- **Report:** every model pair, level and period; the number of tail days behind each comparison
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

What the Researcher Must Check

- **Sign and level convention:** $\text{VaR } 1\% = -q_{1\%}$, positive for a loss; an assistant may label the level by its confidence or flip signs $\Rightarrow$ recompute one day of FZ0 step by step
- **Look-ahead:** GARCH parameters or residual quantiles from the full sample $\Rightarrow$ assert that each forecast uses only data before its day
- **Non-elicitable target:** ranking ES forecasts by mean squared error against losses is not consistent [Gneiting \(2011\)](#) $\Rightarrow$ joint scores for (VaR, ES) only [Fissler & Ziegel \(2016\)](#)
- **Power:** about $n\alpha$ tail days drive the score $\Rightarrow$ count them and report subperiods
- **Multiple testing:** models $\times$ levels $\times$ periods $\times$ assets $\Rightarrow$ fix the family in advance and adjust the p -values
- **Invented references:** an unresolvable DOI or a misattributed score $\Rightarrow$ drop the claim and log it in `AI_ERRORS.md`

Mini-Case: Can We Rank ES Models?

- A typical assistant proposal: “conditional EVT is the best (VaR 1%, ES 1%) model for the S&P 500”; $H_0: \mathbb{E}[d_t] = 0$
- Replicate: FHS exceedance rate 1.54% (80 of 5 210 days, 2006–2026)
- Mean FZ0 (lower is better): conditional EVT 1.267, FHS 1.283, GARCH with Normal shocks 1.453
 - ▶ FHS vs GARCH-Normal: $\bar{d} = -0.170$, $t = -4.48$; $t = -3.24$ in 2006–2015 and -3.38 in 2016–2026
 - ▶ Conditional EVT vs FHS: $\bar{d} = -0.016$, $t = -2.62$ (two-sided $p = 0.009$, unadjusted); $t = -2.31$ in 2006–2015 but -1.23 in 2016–2026
- The shape of the shock distribution matters robustly: Normal shocks lose in both halves
- The EVT gain over FHS is about ten times smaller: significant (unadjusted) on the full sample and in 2006–2015, not in 2016–2026, where the 95% interval for $\mathbb{E}[d_t]$ is $[-0.018, 0.004]$: the recent advantage is not established, which is not proof of equivalence

Your Turn: A Team-Project Seed

- Question: does a dynamic semiparametric (VaR, ES) model [Patton, Ziegel & Chen \(2019\)](#) beat FHS and conditional EVT at ES 2.5% on the S&P 500, the BET and Bitcoin, out of sample?
- Replicate first: the conditional numbers of Section 4 (S&P 500 FHS VaR 1% 1.97%) and the FZ0 ranking of the mini-case
- Deliverables
 - ▶ A one-page pre-registration: models, levels, scores, subperiods, power (tail days) and what would falsify the claim
 - ▶ Code on data/market, a table of all comparisons with adjusted p -values, a referee-style self-critique
- Log every prompt, tool and accepted output in `AI_USE.md`; every error caught (flipped sign, wrong level label, look-ahead, invented DOI) and the test that caught it in `AI_ERRORS.md`
- Code of the mini-case: [Quantlets/Ch_07/ai_discovery_case.py](#)

Summary

- ▣ VaR is a quantile of the loss, ES the average beyond it and a convex minimum (Rockafellar–Uryasev); ES is coherent, VaR is not in general
- ▣ Every VaR or ES is an estimate: quantile and ES asymptotics, HAC or bootstrap intervals, estimation risk in conditional VaR
- ▣ Heavy tails: the Normal distribution underestimates, Cornish–Fisher overshoots, Student- t and EVT fit well
- ▣ Volatility clustering: filtered historical simulation and conditional EVT react in days, not months
- ▣ $\sqrt{h}$ scaling fails with autocorrelation, clustering and heavy tails; simulate instead
- ▣ Euler allocation shows where the risk sits; the 2019 Basel standard (FRTB) sets internal-model capital by ES 2.5%, in the EU from 1 January 2027

Which Method When?

Situation	Suggested method
Quick check, long stable history	Historical simulation, 500–1000 days
Daily risk after a volatility shock	Filtered historical simulation or conditional EVT
VaR 0.5% and deeper in the tail, stress tests	Peaks over threshold (GPD), conditional or not
Multi-day horizon, options, non-linear pay-offs	Filtered Monte Carlo
Risk budgeting across desks or assets	Euler contributions to ES
Communication with a regulator	ES 2.5%, 10 days, stressed period, plus VaR 1% backtests

Coming Next: Backtesting (Chapter 8)

- ▣ How often did losses exceed VaR, and were the exceedances independent?
- ▣ Unconditional and conditional coverage tests; the Basel traffic-light zones
- ▣ Backtesting ES: joint scoring functions for (VaR, ES) Fissler & Ziegel (2016)
- ▣ Further reading: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), McNeil, Frey & Embrechts (2015)

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Measuring Statistical Risk
- ▣  XFG Advanced Methods in Quantitative Finance

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_07

References (1/6)

- ▣ Acerbi, C., Tasche, D. (2002). [On the coherence of expected shortfall](#). *Journal of Banking & Finance*, 26(7), 1487–1503.
- ▣ Artzner, P., Delbaen, F., Eber, J.-M., Heath, D. (1999). [Coherent measures of risk](#). *Mathematical Finance*, 9(3), 203–228.
- ▣ Balkema, A.A., de Haan, L. (1974). [Residual life time at great age](#). *Annals of Probability*, 2(5), 792–804.
- ▣ Bangia, A., Diebold, F.X., Schuermann, T., Stroughair, J.D. (2001). [Modeling liquidity risk, with implications for traditional market risk measurement and management](#). In *Risk Management: The State of the Art*, Springer, 3–13.
- ▣ Barone-Adesi, G., Giannopoulos, K., Vosper, L. (1999). [VaR without correlations for portfolios of derivative securities](#). *Journal of Futures Markets*, 19(5), 583–602.
- ▣ Basel Committee on Banking Supervision (1996). [Amendment to the capital accord to incorporate market risks](#). Bank for International Settlements.
- ▣ Basel Committee on Banking Supervision (2019). [Minimum capital requirements for market risk](#). Bank for International Settlements.
- ▣ Bayer, S., Dimitriadis, T. (2022). [Regression-based expected shortfall backtesting](#). *Journal of Financial Econometrics*, 20(3), 437–471.
- ▣ Bollerslev, T. (1986). [Generalized autoregressive conditional heteroskedasticity](#). *Journal of Econometrics*, 31(3), 307–327.
- ▣ Chen, S.X. (2008). [Nonparametric estimation of expected shortfall](#). *Journal of Financial Econometrics*, 6(1), 87–107.

References (2/6)

- ▣ Christoffersen, P., Gonçalves, S. (2005). Estimation risk in financial risk management. *Journal of Risk*, 7(3), 1–28.
- ▣ Čížek, P., Härdle, W.K., Weron, R. (Eds.) (2011). *Statistical Tools for Finance and Insurance*, 2nd ed. Springer.
- ▣ Cont, R., Deguest, R., Scandolo, G. (2010). Robustness and sensitivity analysis of risk measurement procedures. *Quantitative Finance*, 10(6), 593–606.
- ▣ Cornish, E.A., Fisher, R.A. (1938). Moments and cumulants in the specification of distributions. *Revue de l'Institut International de Statistique*, 5(4), 307–320.
- ▣ Daniélsson, J. (2011). *Financial Risk Forecasting*. Wiley.
- ▣ Daniélsson, J., Jorgensen, B.N., Samorodnitsky, G., Sarma, M., de Vries, C.G. (2013). Fat tails, VaR and subadditivity. *Journal of Econometrics*, 172(2), 283–291.
- ▣ Daniélsson, J., Zigrand, J.-P. (2006). On time-scaling of risk and the square-root-of-time rule. *Journal of Banking & Finance*, 30(10), 2701–2713.
- ▣ Dimitriadis, T., Bayer, S. (2019). A joint quantile and expected shortfall regression framework. *Electronic Journal of Statistics*, 13(1), 1823–1871.
- ▣ Embrechts, P., Klüppelberg, C., Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer.
- ▣ Embrechts, P., Puccetti, G., Rüschendorf, L. (2013). Model uncertainty and VaR aggregation. *Journal of Banking & Finance*, 37(8), 2750–2764.

References (3/6)

- ▣ Engle, R.F., Manganelli, S. (2004). [CAViaR: conditional autoregressive value at risk by regression quantiles](#). *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Ferro, C.A.T., Segers, J. (2003). [Inference for clusters of extreme values](#). *Journal of the Royal Statistical Society: Series B*, 65(2), 545–556.
- ▣ Fisher, R.A., Tippett, L.H.C. (1928). [Limiting forms of the frequency distribution of the largest or smallest member of a sample](#). *Mathematical Proceedings of the Cambridge Philosophical Society*, 24(2), 180–190.
- ▣ Fissler, T., Ziegel, J.F. (2016). [Higher order elicibility and Osband's principle](#). *Annals of Statistics*, 44(4), 1680–1707.
- ▣ Francq, C., Zakoïan, J.-M. (2015). [Risk-parameter estimation in volatility models](#). *Journal of Econometrics*, 184(1), 158–173.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Gnedenko, B. (1943). [Sur la distribution limite du terme maximum d'une série aléatoire](#). *Annals of Mathematics*, 44(3), 423–453.
- ▣ Gneiting, T. (2011). [Making and evaluating point forecasts](#). *Journal of the American Statistical Association*, 106(494), 746–762.
- ▣ Gumbel, E.J. (1958). [Statistics of Extremes](#). Columbia University Press.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). [Applied Quantitative Finance](#), 3rd ed. Springer.

References (4/6)

- ▣ Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- ▣ Hull, J., White, A. (1998). Incorporating volatility updating into the historical simulation method for value-at-risk. *Journal of Risk*, 1(1), 5–19.
- ▣ J.P. Morgan/Reuters (1996). *RiskMetrics – Technical Document*, 4th ed. New York.
- ▣ Koenker, R., Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1), 33–50.
- ▣ Longin, F.M. (2000). From value at risk to stress testing: the extreme value approach. *Journal of Banking & Finance*, 24(7), 1097–1130.
- ▣ McNeil, A.J., Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). *Quantitative Risk Management: Concepts, Techniques and Tools*, revised ed. Princeton University Press.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Newey, W.K., West, K.D. (1994). Automatic lag selection in covariance matrix estimation. *Review of Economic Studies*, 61(4), 631–653.
- ▣ Nolde, N., Ziegel, J.F. (2017). Elicitability and backtesting: perspectives for banking regulation. *Annals of Applied Statistics*, 11(4), 1833–1874.

References (5/6)

- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and Value-at-Risk\)](#). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). [Information entropy and measures of market risk](#). *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Lazar, E., Mazurencu-Marinescu-Pele, M. (2019). [Modeling expected shortfall using tail entropy](#). *Entropy*, 21(12), 1204.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). [Using high-frequency entropy to forecast Bitcoin's daily value at risk](#). *Entropy*, 21(2), 102.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). [Finite-sample precision limits for expected shortfall forecast comparisons](#). *Mathematics*, 14(13), 2316.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Pickands, J. (1975). [Statistical inference using extreme order statistics](#). *Annals of Statistics*, 3(1), 119–131.
- ▣ Rockafellar, R.T., Uryasev, S. (2000). [Optimization of conditional value-at-risk](#). *Journal of Risk*, 2(3), 21–41.
- ▣ Rockafellar, R.T., Uryasev, S. (2002). [Conditional value-at-risk for general loss distributions](#). *Journal of Banking & Finance*, 26(7), 1443–1471.
- ▣ Spierdijk, L. (2016). [Confidence intervals for ARMA–GARCH value-at-risk: the case of heavy tails and skewness](#). *Computational Statistics & Data Analysis*, 100, 545–559.

References (6/6)

- ▣ Tasche, D. (2008). Capital allocation to business units and sub-portfolios: the Euler principle. [arXiv:0708.2542](#).
- ▣ Taylor, J.W. (2019). Forecasting value at risk and expected shortfall using a semiparametric approach based on the asymmetric Laplace distribution. *Journal of Business & Economic Statistics*, 37(1), 121–133.
- ▣ Yamai, Y., Yoshida, T. (2005). Value-at-risk versus expected shortfall: a practical perspective. *Journal of Banking & Finance*, 29(4), 997–1015.